

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I __Dinesh S(192521152)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Cloud computing

Cloud computing is the on demand delivery of computing such as servers, storage, databases, networking and software over the internet with pay as you use pricing.

Characteristics:

- 1) On demand self service: users can access resources without human intervention.
- 2) Broad network access: services are available over the internet on different device.
- 3) Resource pooling: Resources are shared among multiple users using virtualization.
- 4) Measured services: users pay only for resources they consume.

2. Evolution of cloud computing

- * Hardware evolution → Mainframes → personal computers → client-server system
- * Networking: High speed internet enables a single physical server
- * Virtualization: Multiple virtual machine run on a global connectivity
- * Web technologies: web application and

web services becomes common

* Modern cloud: providers like AWS, Azure and Google cloud offers scalable and on-demand service

Therefore: Faster deployment, lower cost, high scalable and global access

3. Service virtualisation

It is the process of creating multiple virtual servers on a single physical server using a hypervisor.

Roles:

- * Improves hardware virtualisation
- * Reduces cost
- * supports easy management and isolation

Virtualization	cloud computing
Technology	service model
creates virtual machine	Delivers service over the internet
Foundation of cloud	Built using virtualisation

4. Data centre:

Data center is the facility that contain servers, storage system, Networking Equipment and security system.

Roles:

- * Store user Data securely
- * Host cloud application and website
- * provide high availability and backup system
- * Ensure fast processing through the powerful server.
- * Supports disaster Recovery and Business
- * Maintain security using firewall and monitoring system

Data centers is the backbone of cloud computing because they provide reliable and security system.

5. Differentiate IaaS, PaaS, SaaS and XaaS with example.

	Description	Example
IaaS	Provide virtual server storage and Networking	Amazon EC2
PaaS	Provide platform to develop and Deploy Application	google APP
SaaS	Provide software application through internet	gmail
XaaS	"Anything as a service" offers various IT services	Microsoft Azure service